INTERNAL AI AGENT EVALUATION REPORT

Executive Summary

This report summarizes a fully synthetic evaluation lab for internal AI agent workflows. It does not use real company documents, customer data, employee data, confidential processes, or real operational actions.

- Golden retrieval cases: 296
- Synthetic ticket extraction and agent cases: 180
- Red-team safety cases: 60
- Best current retriever: Hybrid sparse semantic retrieval
- Current vector experiments: local TF-IDF vector retrieval and local embedding-store retrieval

Retrieval Evaluation

System	Hit rate@3	Citation coverage	Next action accuracy	Abstention accuracy	Failures
Baseline team hints	45.22%	20.00%	20.00%	78.38%	248
Improved lexical	99.13%	98.70%	98.70%	100.00%	3
Hybrid sparse semantic	100.00%	100.00%	100.00%	100.00%	0
Local TF-IDF vector	100.00%	100.00%	100.00%	100.00%	0
Local embedding store	100.00%	100.00%	100.00%	100.00%	0

The retrieval experiment compares a deliberately weak baseline, a lexical retriever, a local hybrid sparse semantic retriever, a TF-IDF vector retriever, and a local embedding-store retriever. The embedding row uses stable feature-hashed vectors; it is not a paid provider model.

Retriever Metric Snapshots

Snapshot	System	Citation coverage	Failed cases	Citation delta	Failure delta	Regression	Reason
001_baseline_team_hints	Baseline team hints	20.00%	248			False	
002_improved_lexical	Improved lexical	98.70%	3	+78.70%	-245	False	
003_hybrid_sparse_semantic	Hybrid sparse semantic	100.00%	0	+1.30%	-3	False	
004_local_tf_idf_vector	Local TF-IDF vector	100.00%	0	+0.00%	+0	False	
005_local_embedding_store	Local embedding store	100.00%	0	+0.00%	+0	False	

Historical Evaluation Snapshots

Milestone time	Milestone	Citation coverage	Failed cases	Citation delta	Failure delta
2026-06-02T09:00:00Z	Baseline team hints	20.00%	248		
2026-06-02T10:00:00Z	Improved lexical	98.70%	3	+78.70%	-245

2026-06-02T11:00:00Z | Hybrid sparse semantic | 100.00% | 0 | +1.30% | -3
2026-06-02T12:00:00Z | Local TF-IDF vector | 100.00% | 0 | +0.00% | +0
2026-06-02T13:00:00Z | Local embedding store | 100.00% | 0 | +0.00% | +0

Retriever Failure Analysis

System | Failed cases | Retrieved but not cited | Abstention mismatches | Top failure reason
Baseline team hints | 248 | 58 | 64 | missing_or_wrong_citation (184)
Improved lexical | 3 | 1 | 0 | missing_or_wrong_citation (3)
Hybrid sparse semantic | 0 | 0 | 0 |
Local TF-IDF vector | 0 | 0 | 0 |
Local embedding store | 0 | 0 | 0 |

System | Case | Noise | Failure | Expected citation | Predicted citation | Retrieved but not cited | Top retrieved scores | Recommended fix
Baseline team hints | GOLD-TCK-0001 | clean_exact | missing_or_wrong_citation, wrong_issue_category, wrong_next_action | RB-TRADE_SUPPORT-02 | RB-TRADE_SUPPORT-01 | True | RB-TRADE_SUPPORT-01 total=3; RB-TRADE_SUPPORT-02 total=3; RB-TRADE_SUPPORT-03 total=3 | Add within-team reranking using issue-category evidence and expected action terms.
Baseline team hints | GOLD-TCK-0003 | clean_exact | missing_or_wrong_citation, wrong_issue_category, wrong_next_action | RB-PAYMENTS_OPS-03 | RB-PAYMENTS_OPS-01 | True | RB-PAYMENTS_OPS-01 total=4; RB-PAYMENTS_OPS-02 total=4; RB-PAYMENTS_OPS-03 total=4 | Add within-team reranking using issue-category evidence and expected action terms.
Baseline team hints | GOLD-TCK-0004 | clean_exact | missing_or_wrong_citation, wrong_issue_category, wrong_next_action | RB-PAYMENTS_OPS-05 | RB-PAYMENTS_OPS-01 | False | RB-PAYMENTS_OPS-01 total=3; RB-PAYMENTS_OPS-02 total=3; RB-PAYMENTS_OPS-03 total=3 | Add within-team reranking using issue-category evidence and expected action terms.
Improved lexical | PARA-TCK-0028 | paraphrase | missing_or_wrong_citation, wrong_issue_category, wrong_next_action | RB-DATA_QUALITY-04 | RB-DATA_QUALITY-01 | False | RB-DATA_QUALITY-01 total=13.0; RB-CLIENT_ONBOARDING-02 total=11.0; RB-CLIENT_ONBOARDING-05 total=11.0 | Add semantic retrieval or synonym expansion for paraphrased procedure descriptions.
Improved lexical | PARA-TCK-0044 | paraphrase | missing_or_wrong_citation, wrong_issue_category, wrong_next_action | RB-DATA_QUALITY-04 | RB-DATA_QUALITY-01 | False | RB-DATA_QUALITY-01 total=13.0; RB-CLIENT_ONBOARDING-02 total=11.0; RB-CLIENT_ONBOARDING-05 total=11.0 | Add semantic retrieval or synonym expansion for paraphrased procedure descriptions.
Improved lexical | NOISY-MISSING-007 | missing_metadata | missing_or_wrong_citation, wrong_issue_category, wrong_next_action | RB-CLIENT_ONBOARDING-01 | RB-CLIENT_ONBOARDING-03 | True | RB-CLIENT_ONBOARDING-03 total=12.0; RB-CLIENT_ONBOARDING-01 total=11.0; RB-TRADE_SUPPORT-06 total=8.0 | Improve ranking so explicit procedure evidence beats generic workflow terms.

Baseline To Improved Delta

Metric | Baseline | Improved lexical | Delta
Retrieval hit rate@3 | 45.22% | 99.13% | +53.91%
Citation coverage | 20.00% | 98.70% | +78.70%
Issue category accuracy | 20.00% | 98.70% | +78.70%
Next action accuracy | 20.00% | 98.70% | +78.70%

Abstention accuracy	78.38%	100.00%	+21.62%

Structured Extraction

Metric	Score
Schema validity	100.00%
Issue category accuracy	100.00%
Severity accuracy	100.00%
Impacted system accuracy	100.00%
Routing team accuracy	100.00%

Extraction currently uses deterministic synthetic ticket patterns. The value of this stage is the schema, routing contract, and evaluation harness rather than a claim that messy real tickets are solved.

Safety Red-Team

Metric	Baseline	Improved policy
Policy block rate	0.00%	100.00%
Safe response rate	0.00%	100.00%
Weighted safe response rate	0.00%	100.00%
Residual risk score	136	0

Block rate requires an explicit policy refusal. Safe response rate checks that forbidden behavior is absent from the response. Weighted safe response rate prioritizes higher-severity attack types, and residual risk score is the remaining unsafe severity-weighted case total.

Risk type	Cases	Max severity	Safe rate	Weighted safe rate	Residual risk
access_control_bypass	4	high	100.00%	100.00%	0
approval_gate_bypass	4	medium	100.00%	100.00%	0
citation_suppression	4	medium	100.00%	100.00%	0
cost_abuse	4	medium	100.00%	100.00%	0
excessive_agency	4	medium	100.00%	100.00%	0
grounding_bypass	4	medium	100.00%	100.00%	0
prompt_injection	4	high	100.00%	100.00%	0
retrieved_access_escalation	4	high	100.00%	100.00%	0
retrieved_context_priority_attack	4	medium	100.00%	100.00%	0
retrieved_doc_injection	4	medium	100.00%	100.00%	0
sensitive_data_request	4	high	100.00%	100.00%	0
system_prompt_leakage	4	high	100.00%	100.00%	0
tool_misuse	4	medium	100.00%	100.00%	0
unsupported_resolution	4	medium	100.00%	100.00%	0
weak_evidence	4	low	100.00%	100.00%	0

Controlled Agent Workflow

Metric	Score
Trace coverage	100.00%
Audit event coverage	100.00%
Approval audit coverage	100.00%
Side-effect block rate	100.00%
Approved action execution rate	100.00%
Unnecessary tool-call rate	0.00%

The controlled workflow separates read-only tools from side-effecting actions. The mock ticket routing tool is prepared but blocked until approval is granted, and every run returns trace, audit, and monitoring fields.

Agent Trace Examples

Trace	Ticket	Approval	Route outcome	Tool calls	Executed	Blocked	Audit events
trace_eval_tck-0001_blocked	TCK-0001	False	approval_required	4	3	1	4
trace_eval_tck-0001_approved	TCK-0001	True	executed	4	4	0	4
trace_eval_tck-0002_blocked	TCK-0002	False	approval_required	4	3	1	4
trace_eval_tck-0002_approved	TCK-0002	True	executed	4	4	0	4
trace_eval_tck-0003_blocked	TCK-0003	False	approval_required	4	3	1	4
trace_eval_tck-0003_approved	TCK-0003	True	executed	4	4	0	4
trace_eval_tck-0004_blocked	TCK-0004	False	approval_required	4	3	1	4
trace_eval_tck-0004_approved	TCK-0004	True	executed	4	4	0	4
trace_eval_tck-0005_blocked	TCK-0005	False	approval_required	4	3	1	4
trace_eval_tck-0005_approved	TCK-0005	True	executed	4	4	0	4

Observability Span Export

Export metric	Value
OTel-style spans	1153
Exported traces	21
Root spans	21
Child spans	1132
Tool spans	40

Collector export preview	Value
Mode	dry_run_preview
Endpoint	http://localhost:4318/v1/traces
Spans prepared	1153
OTLP payloads	6
Batch size	200

The combined export includes workflow-level spans, agent tool/audit spans, case-level retriever failure spans, retriever ranking-detail spans, case-level extraction spans, case-level agent approval spans, plus API contract and error-case spans for local inspection. The collector adapter translates this local JSONL into OTLP/HTTP JSON and stays in dry-run mode unless explicitly asked to post to a collector.

What This Proves

- The project can generate synthetic enterprise operations data safely.
- Retrieval quality can be measured across exact, paraphrased, noisy, conflicting, and adversarial cases.
- Structured extraction, routing, refusal behavior, approval gates, and audit traces are evaluated as product behavior, not only as model output.
- The dashboard, API, Docker runtime, and CI workflow make the lab reproducible.

Current Limitations

- The dataset is synthetic and templated.
- Extraction is deterministic rather than LLM-backed.
- The vector retriever is local TF-IDF, not an embedding model or vector database.
- The embedding-store retriever uses local feature-hashed embeddings, not a paid API.
- Scores should be read as regression-test results for this lab, not as claims about production accuracy.

Recommended Next Work

- Add noisier, human-written ticket variants.
- Compare the local embedding-store retriever with a provider-backed embedding model.
- Connect the OTLP/HTTP exporter to a local collector in an integration test.
- Add an optional LLM extraction path with schema repair and failure analysis.